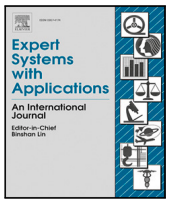
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A higher-order neural cognitive diagnosis model with hierarchical attention networks

Tao Huang [a,d], Yuxia Chen [a], Jing Geng [b,*], Huali Yang [c], Shengze Hu [a]

[a] National Engineering Research Center of Educational Big Data, Central China Normal University, Wuhan, 430079, China
[b] School of Education, Jianghan University, Wuhan, 430056, China
[c] School of Computer Science and Artificial Intelligence, Wuhan Textile University, Wuhan, 430200, China
[d] Ningxia Normal University, Ningxia, 756000, China

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ABSTRACT

As fundamental abilities, higher-order abilities play an important role in representing a multidimensional synthesis of concepts, skills, and learning statuses. Cognitive diagnosis, a critical technology for assessing these abilities, aims to identify students' specific skill statuses based on their response data. However, existing cognitive diagnostic models (CDMs) face significant challenges in accurately assessing these complex competencies, particularly in capturing the hierarchical structure of higher-order abilities—specifically, the progression from lower- to higher-order skills. To address this challenge, we propose a novel Higher-Order Neural Cognitive Diagnosis (HO-NCD) model, which leverages hierarchical attention mechanisms to assess higher-order abilities. The core of our model lies in its ability to model the transition from lower- to higher-order abilities, capturing both the intrinsic relationships between different levels of cognitive attributes. Specifically, a unified embedding layer encodes the characteristics of students, exercises, and concepts in a shared latent space, enabling a comprehensive representation. Subsequently, we model the transition from lower- to higher-order abilities through a hierarchical attention network, allowing the model to capture both direct and indirect relationships. Finally, a neural network is constructed to simulate the interactions among students, exercises, and concepts, thereby predicting future student performance. The effectiveness and interpretability of the HO-NCD model were evaluated using three benchmark datasets — Junyi, ASSIST2017, and PISA2015 — demonstrating its superior performance compared to existing models. The code is available at https://github.com/ccc-615/HO-NCD.

1. Introduction

In the context of digital education, higher-order abilities continue to be essential competencies in the modern paradigm (Brookhart, 2010; Lu et al., 2021). These abilities extend beyond the mere application of basic knowledge, encompassing dynamic and complex learning processes that are crucial for addressing real-world challenges (Arter & Salmon, 1987; Miterianifa et al., 2021). Despite recent advancements in educational technology, traditional methods of assessing these abilities have significant limitations, highlighting the need for innovative and adaptive diagnostic frameworks (Saul & Wuttke, 2011). Cognitive Diagnosis Assessment (CDA) is a critical tool for interpreting students' learning states by evaluating their proficiency in specific cognitive domains (Huang, Geng, Yang, Hu, Ou et al., 2024). An illustrative example of CDA is shown in Fig. 1(a). Generally, by analyzing students' responses, we can assess their proficiency levels in relation to specific concepts. Subsequently, using cognitive diagnostic tools, we may explore the relationship between students' proficiency and the concepts examined during exercises. Accordingly, cognitive diagnoses focusing on the assessment of higher-order abilities have become a prominent research topic.

Existing cognitive diagnostic models (CDMs) typically focus on either micro-level cognitive processes or macro-level abilities in isolation (Gao et al., 2019; Zhang et al., 2024). To assess students' macro-level abilities, Item Response Theory (IRT) (Lord, 2012) was proposed as a mathematical model for analyzing exercise scores and questionnaire data. Several models have subsequently been developed from IRT, including the two-parameter logistic model (2PL-IRT) (Lord & Novick, 2008) and Multidimensional Item Response Theory (MIRT) (Ackerman et al., 2003). In contrast, researchers investigating micro-level cognitive

* Corresponding author.
E-mail addresses: tmht@mail.ccnu.edu.cn (T. Huang), chenyx0615@mails.ccnu.edu.cn (Y. Chen), gengjing@mails.ccnu.edu.cn (J. Geng), yanghuali@mail.ccnu.edu.cn (H. Yang), hsz666@mails.ccnu.edu.cn (S. Hu).

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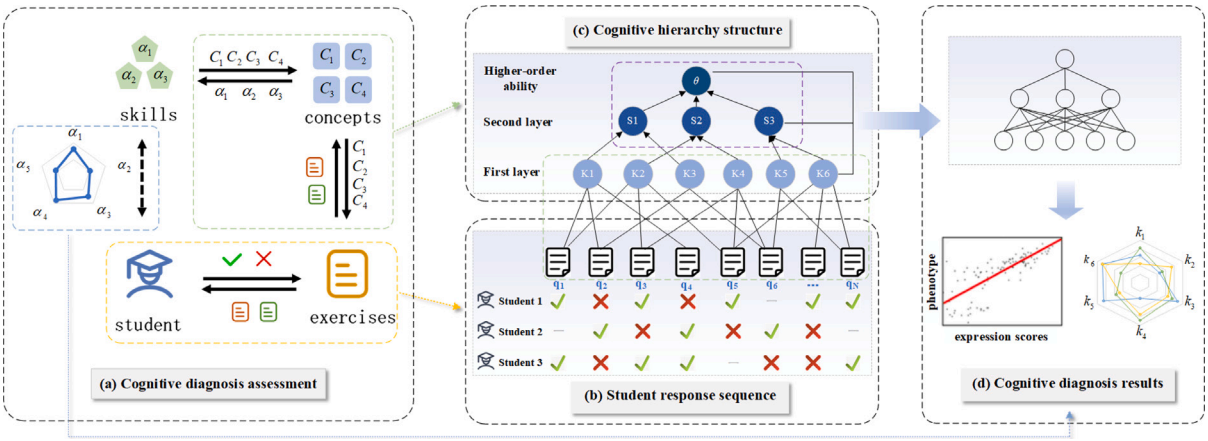


Fig. 1. A toy example of cognitive diagnosis. (a) Students' cognitive diagnosis assessment process; (b) response sequence of students; (c) cognitive hierarchy structure; (d) the results of cognitive states diagnosis and predictions.

processes have introduced the Deterministic Inputs, Noisy 'And' Gate model (DINA) (De La Torre, 2009) to evaluate students' mastery of specific concepts. As shown in Fig. 1(d). As neural networks continue to evolve (Rodríguez-Hernández et al., 2021), they are increasingly employed to diagnose students' cognitive states and generate predictions. Examples of such networks include the classical Neural Cognitive Diagnosis (Neural CD) model (Wang et al., 2020), Interpretable Cognitive Diagnosis (ICD) model (Qi et al., 2023), and Bayesian-network-based Hierarchical Cognitive Diagnosis Framework (Hier CDF) (Li et al., 2022). However, these models have a limited ability to holistically assess higher-order abilities owing to the challenges of modeling complex interrelations between cognitive attributes and tasks, as well as their reliance on static assumptions and high computational demands (Wu et al., 2022). Consequently, there is an urgent demand for robust, scalable, and interpretable diagnostic models.

Since the introduction of transformer architecture (Subakan et al., 2021), attention mechanisms have become transformative tools in various fields, including educational technology. Their ability to model intricate relationships between multidimensional features has positioned attention mechanisms as a promising cognitive diagnosis technology (Niu et al., 2021). In particular, multi-head attention mechanisms, integrated with feature transfer techniques, have demonstrated significant promise for enhancing predictive performance and generalizability (Xia et al., 2022). Recent advancements include multi-modal attention networks that integrate student behavioral data with concept hierarchies, offering more precise assessments of higher-order abilities (de Santana Correia & Colombini, 2022). For example, some researchers have proposed hierarchical attentional networks to represent the effects of specific situations on students and then used adaptive optimization to characterize these effects, balancing high interpretability with superior performance (Zhou et al., 2021). However, existing attention-based models exhibit several notable limitations. Many of these models focus on static feature learning, overlooking the dynamic and iterative nature of learning processes. Furthermore, they often fail to capture the deep hierarchical relationships between higher-order abilities and task-specific concepts, which are crucial for comprehensive cognitive diagnosis (Song et al., 2023).

To address these issues, we propose a novel higher-order cognitive diagnostic model — the Hierarchical Attention-Network-Based Higher-Order Cognitive Diagnosis (HO-NCD) model — designed to uncover deep and intrinsic relationships between concepts and abilities. As illustrated in Fig. 1(c), we collected student response data shown in Fig. 1(b) and constructed a hierarchical structure to mine deep relationships from lower to higher orders. Furthermore, HO-NCD integrates advanced attention mechanisms to provide a comprehensive evaluation of both lower- and higher-order abilities, enabling a deeper understanding of students' hierarchical cognitive characteristics, as illustrated in Fig. 2, which includes comparisons with existing methods. The contributions of this study are as follows:

- We designed a novel higher-order neural cognitive diagnosis framework based on hierarchical attention networks and investigated the correlations between students' cognitive characteristics. Our findings reveal that more capable students tend to exhibit higher correctness rates.
- We utilized advanced attention mechanisms to model the deep relationships between cognitive attributes and concepts, and employed a neural network to simulate the cognitive processes underlying student responses to exercises. Subsequently, the students' cognitive states were inferred by predicting the outcomes of their responses.
- We conducted extensive experiments on three benchmark datasets — Junyi, ASSIST2017, and PISA2015 — to evaluate the performance of HO-NCD. The results demonstrate that the proposed model outperforms state-of-the-art (SOTA) models in performance prediction and convergence. Moreover, the HO-NCD model demonstrates interpretability through an analysis of monotonicity.

The rest of this paper is organized as follows. Section 2 provides a systematic review of the research related to this study. Section 3 presents the formal problem description of the cognitive diagnosis addressed in this study. Section 4 describes the proposed methodology. Section 5 presents a study of three public data sets and an analysis of the results. Section 6 concludes the paper and discusses future research directions.

2. Related work

Cognitive diagnosis models (CDMs) are designed to capture students' implicit cognitive states, especially their skills and abilities, by modeling their observable response data (Zhang et al., 2024). This section provides a comprehensive review of cognitive diagnosis research. CDMs can be broadly categorized as those based on statistical methods, those based on neural networks, or higher-order CDMs.

2.1. CDMs based on statistical method

Early cognitive diagnostic models were primarily developed based on statistical methods to infer students' cognitive states and skill levels. These models relied on probabilistic approaches that used students' observational data — such as response results — and prior

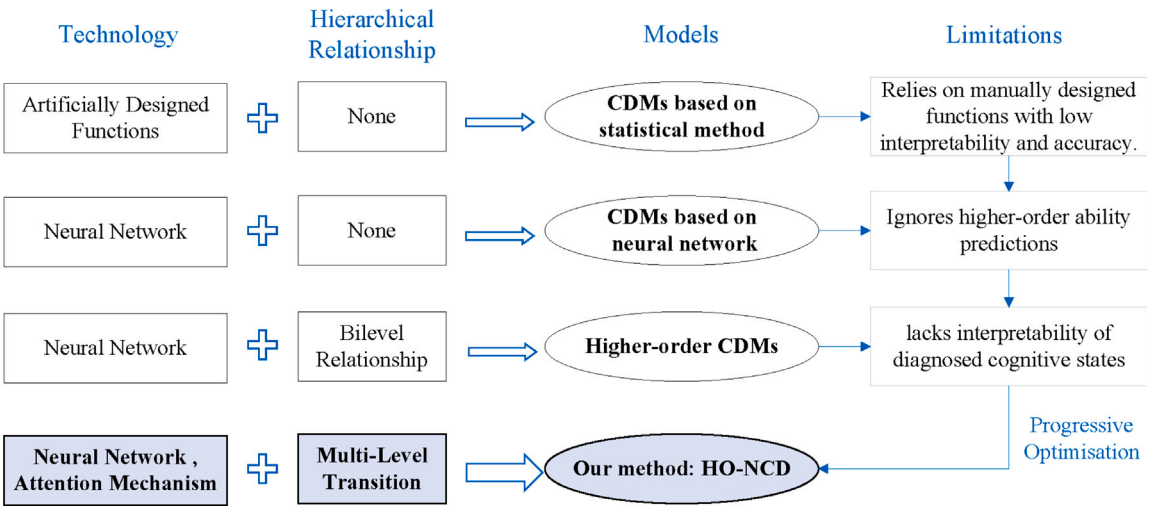


Fig. 2. Comparison of existing models. Early models rely on manual functions, ignoring hierarchies. Neural networks improve modeling but fail to capture hierarchical relationships. Higher-order CDMs mine relationships but lack interpretability. Our model combines neural networks and attention mechanisms to capture ability progression, improving performance and interpretability.

concepts to make inferences. Two such CDMs derived from educational psychology were the unidimensional ability diagnosis approach, represented by Item Response Theory (IRT) (Lord, 2012), and the multidimensional abilities diagnosis approach, represented by the DINA model (De La Torre, 2009). IRT used one-dimensional continuous latent ability θ and exercise difficulty parameter β to represent student attributes and task difficulty, respectively. The interaction between these traits was modeled as $P(\theta) = c + (1-c)\sigma(a(\theta - b))$, where σ was the sigmoid function, c was the guessing parameter of the exercise, and a was the discrimination parameter. In contrast, DINA handled each student's cognitive state as a binary discrete vector, indicating the mastery or non-mastery of relevant concepts. DINA typically assumed non-compensatory mastery, meaning that all required concepts had to be mastered for a correct response. The DINA model was formulated as $\eta_{ij} = \prod_{k=1}^{K} \alpha_{ik}^{q_{jk}}$, where the Q matrix contained q_{jk}, which characterized whether exercise j assessed skill k. The student's skill mastery α_{ik} represented the proficiency of student i in skill k. The latent response value η_{ij} combined the mastery states of all required skills for exercise j and equaled 1 if all necessary skills were mastered; otherwise, it equaled 0. Although certain improvements had been made to traditional CDMs — such as Multidimensional Item Response Theory (MIRT) (Ackerman et al., 2003), which extended IRT by incorporating multidimensional latent features — most item response functions still adhered to logistic formulations (Wang et al., 2020). These conventional models relied heavily on manually designed functions, which limited their flexibility and adaptability (Liu, 2021).

2.2. CDMs based on neural network

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To a certain extent, CDMs based on neural networks relied on the robust learning and fitting capabilities of neural networks (Yang et al., 2023) to mitigate the limitations of statistical-based CDMs in capturing complex interactions between students and exercises. Instead of a manual design, these models directly learned the interaction function from existing data (Ma et al., 2023). Recently, neural CDM (Wang et al., 2020), ICD model (Qi et al., 2023), deep CDM (DCD) (Gao et al., 2022), and multidimensional feature-aware interpretable neurocognitive diagnosis model (MFNCD) (Huang, Geng, Yang, Hu, Ou et al., 2024) emerged as typical CDMs based on neural networks. Neural CDM used multiple neural layers to model students and exercises by considering the intricate relationships between student features and exercise attributes. The specific formula was represented as $Q \circ (h^s - h^{diff})^{disc}$, where h^s denoted student characteristic attributes, h^{diff} denoted practice difficulty attributes, and h^{disc} denoted exercise differentiation. The ICD model employed a new three-layer neural network to capture the relationships between exercise features, effectively differentiating between students at different cognitive levels while maintaining good interpretability. The effect of concepts on exercises was quantitatively represented by the matrix $\mathbf{W} = \frac{q}{1+e^{-w}}$, where q was the relationship between exercises and concepts derived from the Q matrix. A DCD enhanced traditional CDMs, such as IRT or DINA, through deep learning techniques for cognitive diagnosis. This optimized the model and improved interpretability by representing the embedding vectors of exercises and skills according to difficulty and keywords. The MFNCD incorporated temporal features to simultaneously model student response accuracy and speed using neural networks. Based on different hypotheses, researchers developed many CDMs, such as LS-ENCD (Huang, Geng, Yang, Hu, Chen et al., 2024) and SCD (Wang et al., 2023), TechCD (Gao et al., 2023), which became widely used in the field of educational measurement. However, although CDMs based on neural networks overcame some of the limitations of traditional approaches, few studies considered the deep relationships inherent between the concepts assessed through exercises and students' abilities. Consequently, existing models neglected to diagnose and analyze students' higher-order abilities.

As technology continued to evolve, more and more key techniques were applied to cognitive diagnosis models. Attention networks, as the core technology of deep learning, were used in several fields, such as image classification (Guo et al., 2023), semantic segmentation (Xie et al., 2021), and machine translation (Bahdanau, 2014). In recent years, the basic neural network consisted of a series of fully connected layers, each of which produced a hidden representation of fixed dimensions (Wang et al., 2024). Since a hidden layer had to encode all the features of the input, it was difficult to trap the interaction between features. Based on this, more and more researchers investigated CDMs based on attention networks to further enhance the capture of complex features. ECD (Educational context-aware Cognitive Diagnosis) (Zhou et al., 2021) was proposed to optimize the impact of multiple student factors on predicted outcomes. This model used hierarchical attentional networks to represent the effects of specific situations on students and then used adaptive optimization to characterize these effects, balancing high interpretability with superior performance. The Attention Driven Cognitive Diagnosis model (ACD) (Shao et al., 2024) consisted of a three-layer neural network of attentional mechanisms. Each layer of

the network calculated the relationship among students, exercises, and concepts separately to obtain the cognitive state of students. The Long Short-term attentional Cognitive Diagnosis model (LS-ENCD) (Huang, Geng, Yang, Hu, Chen et al., 2024) designed a learning migration module based on an attention mechanism to capture the dependence on historical data, which calculated weights by combining time and mastery states.

However, while CDMs based on neural network overcome some of the limitations of traditional approaches (Wang et al., 2020), the existing study ignores the large amount of information provided by students, which leads to reduced interpretability.

2.3. Higher-order CDMs

In recent years, researchers have made significant advancements in neural-network-based CDMs by considering the complex relationships between students' abilities and the concepts examined through exercises. As an extension of DINA, HO-DINA (De La Torre & Douglas, 2004) was a cognitive diagnostic model used for educational measurement and assessment purposes. It approximated students' proficiency based on their responses to binary items while incorporating higher-order interactions between skills. The HO-NCD accounted for the fact that students may have possessed skill sets not explicitly represented in the program's skill vector, providing a more nuanced understanding of students' abilities by considering the complex relationship between concepts and skills. The fuzzy CDF (Wu et al., 2015) applied fuzzy theory to cognitive diagnosis, thereby resolving issues faced by traditional CDMs in diagnosing subjective exercises. Specifically, it modeled students' mastery α of concepts by combining fuzzy set theory with educational assumptions through the fuzzification of their skill abilities. This was achieved through the following formula: $\alpha = \frac{1}{1+e^{-1.7\alpha(\theta-b)}}$, where θ represented the trait vector of a given student and b denoted the exercise difficulty. On this basis, the fuzzy CDF incorporated both error and guessing factors, achieving excellent interpretability. To effectively handle latent traits with a hierarchical-order structure, HO-IRT (Huang et al., 2013) combined the advantages of MIRT and HO-DINA, enabling the simultaneous estimation of parameter values for students' higher-order latent traits. The corresponding formula was $P(X = 1|\theta,\beta) = c + (1-c)\frac{e^{1.7\alpha(\theta-b)}}{1+e^{1.7\alpha(\theta-b)}}$, where a, b, and c denoted the differentiation, difficulty, and guessing parameters of the exercise, respectively. The HO-IRT offered valuable insights into students' overall abilities to comprehend the dimensions of their subject-specific skills. This approach yielded a more realistic representation and could be adapted to meet the requirements of large-scale testing. Hier-CD (Li et al., 2022) was a higher-order CDM that incorporated a CDM adapter to transform students' cognitive states and exercise characteristics into input features for existing diagnostic models.

2.4. Differences with existing works

CDMs based on statistical methods relied on artificially designed functions that exhibited limitations in addressing student characteristics and predictions. In contrast, neural-network-based CDMs mostly ignored higher-order ability predictions, resulting in limited interpretability of students' diagnosed cognitive states. Thus, although the aforementioned models were designed to predict students' responses and diagnose their characteristics, they disregarded the inherent relationship between students' characteristics and concepts. Consequently, these models were unsuitable for diagnosing students' higher-order potential characteristics. Despite the advances in cognitive diagnosis models, a significant research gap remained: existing models failed to capture the complex, higher-order relationships between students' cognitive abilities and their conceptual understanding. This gap was critical, as it limited the ability of current models to predict and interpret students' latent cognitive potential. To address this issue, we proposed HO-NCD as a novel cognitive diagnosis framework that comprehensively considered the intrinsic relationship between exercises and student characteristics, leveraging the attention network to uncover students' latent abilities. Our model was superior in both accuracy and robustness and addressed the limitations of existing cognitive diagnostic models, as shown in Fig. 3.

3. Preliminaries

The following section presents the relevant mathematical symbols, hypotheses, and theories, along with a formal problem description of cognitive diagnosis. Before defining the problem, we present the notations necessary for the various modules, as listed in Table 1.

3.1. Problem statement

Let $S = \{s_1, s_2, \ldots, s_N\}$ denote the set of N students, $E = \{e_1, e_2, \ldots, e_M\}$ denote the set of M exercises, and $K = \{k_1, k_2, \ldots, k_T\}$ denote the set of T concepts. Each student selects a subset of exercises for practice, and their response log R is denoted as a set of triplets (s, e, r) where $s \in S$, $e \in E$ and r_{ij} is the score that student s_i achieved in exercise e_j. The response process is modeled as $r = F(\omega, \phi)$, where r denotes the result of the student's response, ω denotes the student's parameters (mastery of concepts α, response time t) and ϕ denotes the parameters of the exercise (difficulty b, discrimination $disc$).

Definition 1 (*Student Factor*)**.** Let $\alpha_i = \{\alpha_i^1, \alpha_i^2, \ldots, \alpha_i^k\}$ denote the skill proficiency vector of the student $s_i \in S$, where $\alpha_i^k \in [0, 1]$ represents the skill proficiency of the student s_i in the skill $k_t \in K$. In addition, let v_i denote the workload that students can complete in a suit of time, which is the latent attribute of the students s_i measured by the average response time t_i of the exercise.

Definition 2 (*Exercise Factor*)**.** Let τ_j be the workload attribute of exercise e_j, which represents the level of work required to solve exercise e_j, typically determined by response time and speed calculations.

Definition 3 (*Cognitive Factor*)**.** Let θ be the higher-order abilities such as mathematical literacy, which denotes the proficiency in applying the hypothesized unidimensional construct being measured. In this paper, we employ a three-layer attention neural network to extract latent traits from students at a higher-order level by integrating the concepts assessed in exercises with individual student characteristics.

3.2. Higher-order hierarchical framework

We first introduce the Concepts — Skills — Higher-Order Abilities Hierarchical Framework. Subsequently, we formally define the problem of higher-order cognitive diagnosis based on hierarchical attention networks.

Benjamin Samuel Bloom, an American educational psychologist, classified students' cognitive thinking goals into six levels ranging from low to high and from simple to complex (Seddon, 1978). These six levels can be classified as lower- and higher-order thinking skills. Among these six levels, analyzing, evaluating, and creating fall under the category of higher-order abilities (Ercikan et al., 2011). It is widely acknowledged that a comprehensive assessment of higher-order abilities measures students' lower-level skills and evaluates their utilization and integration of concepts (Schraw & Robinson, 2011), particularly critical thinking and problem-solving capabilities (Widana, 2017). Higher-order abilities can be defined as cognitive and learning processes that surpass basic concepts and skills, encompassing deeper levels of thinking, creativity, and problem solving (Anderson & Krathwohl, 2001), as illustrated in Fig. 4. We conceptualize higher-order abilities as abstractions derived from the mastery of simple concepts and skills, forming a three-layer framework structured as "Concepts — Skills — Higher-Order Abilities".

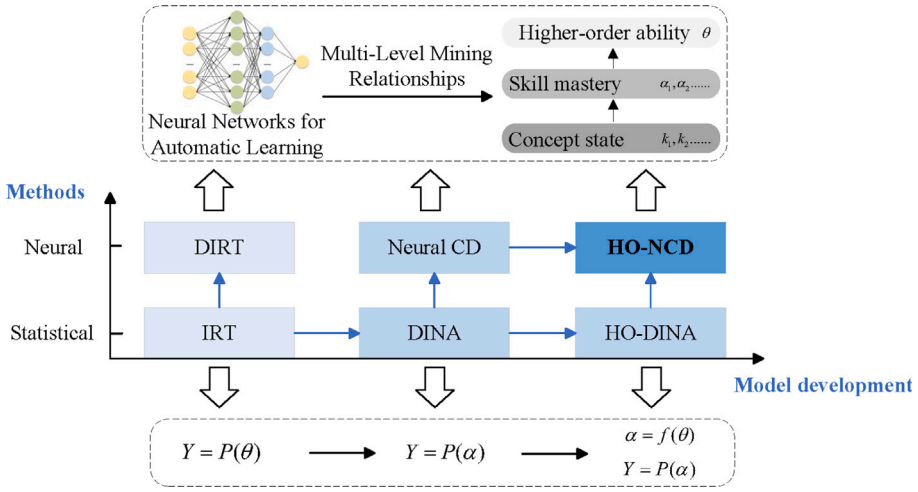


Fig. 3. Differences with existing works. This figure highlights the evolution of cognitive diagnosis models (CDMs), showing how the proposed HO-NCD leverages hierarchical attention mechanisms to address the limitations of traditional and neural models.

Table 1
Important symbols used in this paper.

Symbols	Descriptions
α	The concepts mastery.
b_j	The difficulty attribute of the exercise e_j.
τ_j	Workload attribute of the exercise e_j.
Q	Contained relation matrix between exercises and concepts.
α_i^k	The mastery of skill k of student s_i.
θ	The ability trait of student s_i.
v_i	The response speed of student s_i.
t_i	The response time of student s_i.
x_t	The student characteristic attribute.
x_t'	The final characteristic attributes of students.
w	The match between different concepts and student characteristics (degree of influence).
w_j	The weight vector w value in dimension j.
C_U	The concepts set.
C_{lG}	The concepts set of category l.
σ_l	The student weights for category l concepts.
$\hat{y}_i$	The predicted probability of student s_i to answer correctly.

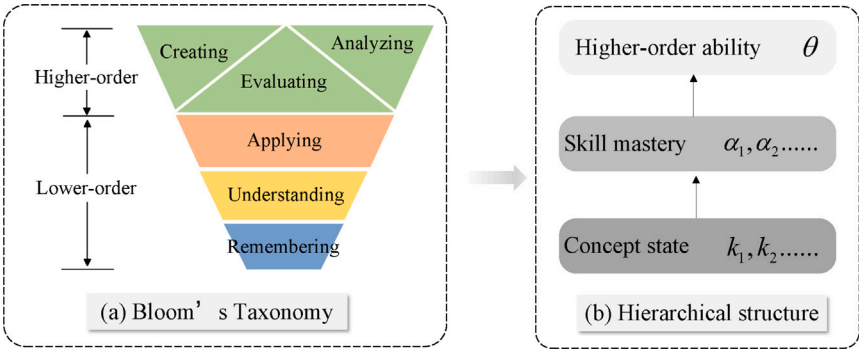


Fig. 4. Higher-order framework. (a) The classification of Bloom's cognitive objectives from lower-order to higher-order; (b) the three-layer of "concepts-skills-higher-order abilities".

We manually classified exercise concepts to obtain the initial representation vector of concepts $C_i(i = 1, 2, \ldots, U)$. These concepts include "equations and inequalities", "probability of random events", and "properties of graphs". Then, by utilizing an attention network, we integrated these concepts with the student's initial characteristics x_t to obtain their mastery $\alpha_i(i = 1, 2, 3 \ldots)$ of skills such as "arithmetic", "reasoning", and "spatial concepts". Finally, we obtained the final higher-order abilities θ (e.g., mathematical literacy) through the interaction between student characteristics and mastery skills.

From a broader perspective, the hierarchical Concepts — Skills — Higher-Order Abilities framework comprises three components, with each element further reinforced by its own dedicated attention network. The relationship between concepts and skills is explored using a self-attention mechanism, as shown in Fig. 5(b). Subsequently, the processed concept features are integrated with the learned weights and computed through a low-order ability attention layer. The final student characteristics are determined by integrating the initial student characteristics, resulting in a comprehensive representation that serves as an input to the diagnostic module. Overall, the three-layer attention network enables the effective capture and assessment of students' higher-order abilities.

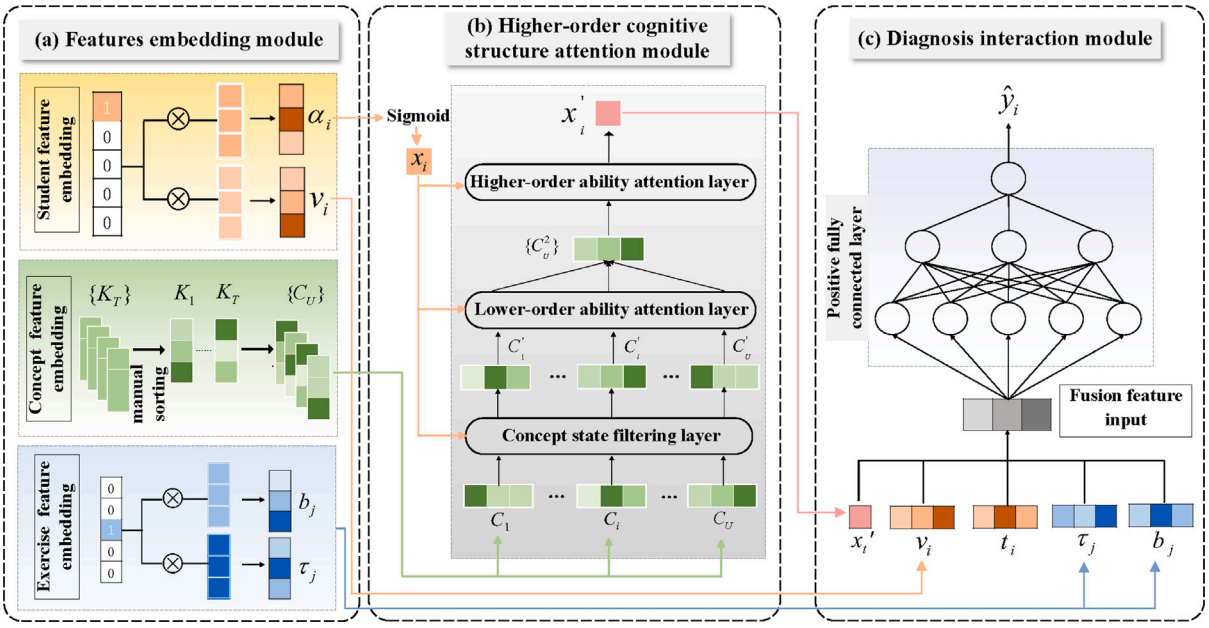


Fig. 5. HO-NCD framework. (a) Embedding modules of students, exercises and concepts features; (b) hierarchical structure from lower-order to higher-order; (c) diagnose and output student performance predictions.

3.3. Problem description

Using preliminary feature embedding, we obtained feature vectors corresponding to students, concepts, and exercises. For each student s_i in the dataset, true scores R and preliminary features are provided through the Q matrix. The three cognitive diagnosis targets are defined as follows:

- **Modeling skill proficiency:** Our objective is to model the skill proficiency vectors of s_i. This involves refining the preliminary feature embeddings to capture the student's final feature, representing their mastery level over various concepts.
- **Fusing factors:** The second objective is to integrate various factors through the function $\theta = \psi(x_i', v_i, t_i, \tau_j, b_j)$. This process aims to compute the cognitive state θ of the student, which serves as a latent representation of them.
- **Score prediction:** Finally, the third objective is to predict the scores of student s_i based on their cognitive state θ. Using the function $P(\hat{y}_i = 1) = \Phi(\theta)$, we estimate $\hat{y}_i$, which can serve as a proxy for predicting student performance in the future.

4. Cognitive diagnosis framework

This section provides a concise overview of the proposed model. As shown in Fig. 5, HO-NCD consists of three main parts: a vector embedding module, a higher-order cognitive structure attention module, and a diagnosis interaction module. The vector embedding module encodes students and exercises in a d-dimensional representation to analyze different concepts. Subsequently, manual classification is employed to cluster similar concepts and reduce their complexity. In the higher-order cognitive structure attention module, an attention mechanism is leveraged to capture the hierarchical relationship between exercise concepts and students' abilities. Each student's characteristic attributes are ultimately generated through a three-layer attention network. Finally, by integrating student attributes with exercise attributes and response times, we can infer students' cognitive levels by predicting their response scores.

4.1. Features embedding module

To derive the student, exercise, and concept factors, we initially embedded the student, exercise, and concepts separately. This process involved encoding them using trainable matrices of dimensionality d.

Student features. First, we combined the student attribute h_i^s with trainable matrices to obtain the student factor x_t using a sigmoid function. As shown in Fig. 5(a), the following equation is used:

$$\alpha_i = h_i^s \times \mathbf{A} \tag{1}$$

$$x_t = \text{sigmoid}(\alpha_i) \tag{2}$$

The response speed v_i is then obtained by combining the student attribute h_i^s with trainable matrices, which are calculated as follows:

$$v_i = h_i^s \times \mathbf{V} \tag{3}$$

where $\mathbf{A}, \mathbf{V} \in \mathbb{R}^{N\times L}$, $h_i^s \in \mathbb{R}^{1\times N}$, $\alpha_i, v_i \in \mathbb{R}^{1\times K}$ are the one-hot representation vectors of s_i respectively, while $\mathbf{A}$ and $\mathbf{V}$ are trainable matrices learned through the data.

Exercise features. For each exercise, the exercise difficulty b_j and differentiation $disc_j$ are respectively multiplied by trainable matrices C and D. That is,

$$b_j = h_j^P \times \mathbf{C} \quad \tau_j = h_j^P \times \mathbf{D} \tag{4}$$

where $\mathbf{C}, \mathbf{D} \in \mathbb{R}^{M\times N}$, $h_j^P \in \mathbb{R}^{1\times M}$, $b_j, \tau_j \in \mathbb{R}^{1\times K}$ is the one-hot representation vectors of e_j respectively, while $\mathbf{C}$ and $\mathbf{D}$ are trainable matrices learned through the data.

Furthermore, owing to the variations in individual student comprehension and response patterns, different concepts are mastered by different students during the same exercises (Yang et al., 2022). Therefore, these concepts must be considered separately. Moreover, as multiple exercises may assess the same concept, students' responses can vary for exercises that evaluate identical concepts.

Concept features. As shown in Fig. 6, we first extract the concept $\{K_T\}$. Subsequently, we employ manual classification to cluster similar concepts within the dataset, resulting in distinct concept types K_i,where $i = 1, 2, \ldots, K$. These types are further categorized, enabling a comprehensive representation of the concepts $C = \{C_1, C_2, \ldots, C_U\}$.

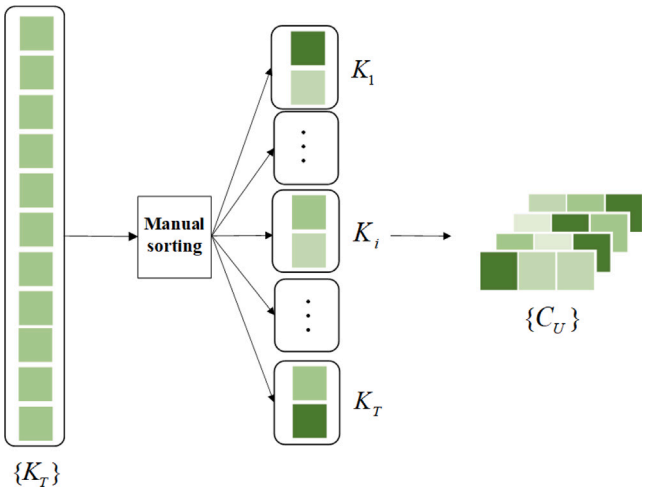


Fig. 6. Embedding module of concept vectors.

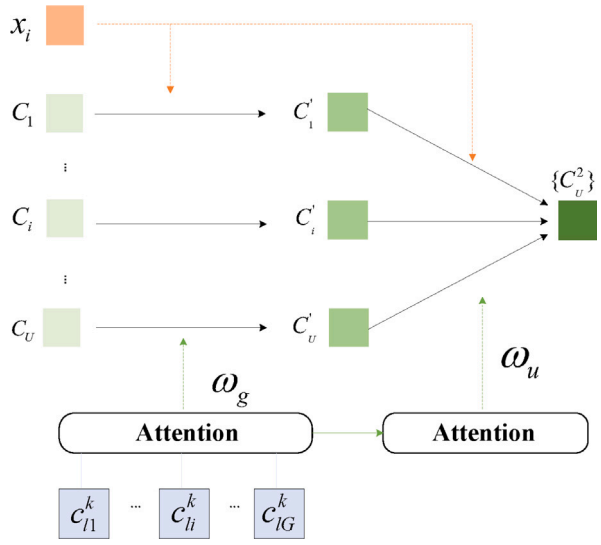


Fig. 7. Concept state filtering and Lower-order ability attention layers.

4.2. Higher-order cognitive structure attention module

This section describes our hierarchical attention network designed to explore the relationship between concepts and skills, thereby facilitating their interactions. The network comprises three layers: concept state filtering, lower-order ability attention, and higher-order ability aggregation.

4.2.1. Concept state filtering layer

After obtaining the vectorized representations of each student s_i and exercise e_j from the previous layer, this layer utilizes the attention mechanism to calculate the relationship between concepts and students' characteristics. Considering that different students may exhibit varying levels of mastery of the same concept and diverse response data for identical exercises, it is unreasonable to model the exercises uniformly. Therefore, we adopted a manual cluster analysis of concepts underlying the different exercises. Specifically, through manual categorization based on similarity, we classify concepts into U distinct categories. Thus, all concepts can be expressed as $C = \{C_1, C_2, \ldots, C_U\}$; furthermore, all concepts in G categories of class l can be expressed as $C = \{C_{l1}, C_{l2}, \ldots, C_{lG}\}$. Subsequently, considering the variations in inner concepts and student mastery levels, an attention mechanism is deployed to compute the weights for different concepts. Unlike traditional approaches that involve summing or averaging concept vectors and student abilities, our method employs an attention mechanism to derive weights. Specifically, we first calculate the cosine similarity between the student characteristic x_t and the concept that influences the key vector $c_l^k = (c_{l1}^k, c_{l2}^k, \ldots c_{lG}^k)$. The concept influence key vector, denoted as c_l^k, represents a learned feature vector that characterizes the influence of different concepts in category l on student performance. These vectors are computed automatically through a neural network that learns to capture the underlying relationship weights between different concepts. The weight is calculated as follows:

$$w = \mathrm{Softmax}(sim(x_t, c_l^k)) \tag{5}$$

where sim represents the similarity function and w denotes the given student's fitness on different concepts. This function essentially measures the angle between these two vectors, and a higher similarity indicates that the student's characteristic aligns more closely with the concept's influence. To ensure the proper use of the sim function, we need to ensure that the input vector dimensions are consistent with the normalized representation. The influence of the potential characteristics of concepts in category l and the representation of the impact on students is generated using the weighted summation of different latent features c_{lG}^k for each concept:

$$\sigma_l = \sum_{g=1}^{G} w_g \times c_{lG}^k \tag{6}$$

where w_g is the gth weight in w, and σ_l represents the effect of category-l concepts on students.

4.2.2. Lower-order ability attention layer

To analyze the relationships between different categories of concepts, this layer incorporates a self-attention mechanism module. As shown in Fig. 7, we obtain a hierarchical relationship between concepts and skills by employing a self-attention mechanism and weight sharing. As mentioned in the previous section, inherent connections or interactions may exist between different categories of concepts. Hence, the self-attention mechanism is used to model the interactions between different categories of concepts. Following the previous layer, we acquire potential information σ_l for all concept categories, where $l \in 1, 2, \ldots, U$. Similarly, we can calculate the mutual influence weights for interactions based on the potential characteristic vectors of the corresponding distinct concepts. This equation is expressed as follows:

$$w = Softmax(sim(\sigma_l, \sigma)) \quad l \in 1, 2, \ldots, U \tag{7}$$

where $\sigma = (\sigma_1, \sigma_2, \ldots, \sigma_U)$.

Following the interaction, the latent characteristics of concepts in category l are generated with a weighted summation between the latent characteristic representation of each type of concept and the student weights:

$$\sigma'_l = \sum_{u=1}^{U} w_u \times \sigma_u \tag{8}$$

where w_u is the uth weight in w and σ'_l is the representation of the impact on students of concepts in category l.

4.2.3. Higher-order ability aggregation layer

After computing the preceding two layers of the network, we obtain the intrinsic relationship between students' abilities and concepts, as shown in the higher-order cognitive structure attention module in Fig. 5(b). This layer employs another attention module to calculate the relationship between different hierarchies: the higher-order hierarchical relationship between skills and higher-order abilities. For example, distinct concepts correspond to the same skill, whereas similar concepts reflect diverse skills. We then obtain the students' final characteristic

attributes. To adaptively learn the weights for different categories of concepts, we utilize an attention mechanism. Similarly, a given student's ultimate characteristic attributes can be computed using the following formula:

$$w' = Softmax(sim(x_t, \sigma_l)) \quad \sigma_l' = (\sigma_1, \sigma_2, \ldots, \sigma_U) \tag{9}$$

$$x_t' = \sum_{l=1}^{U} w' \times \sigma_l' \tag{10}$$

Notably, although the previous two layers employ the attention mechanism, there are disparities in their inputs, and they do not yield identical effects. In the higher-order ability aggregation layer, the inputs encompass diverse categories of concepts rather than individual concepts with student weight representations. However, after traversing through the previous two layers of the network, the representation of potential concept characteristics obtained for each category already considers their interplay and idiosyncrasies. Therefore, in the higher-order ability aggregation layer, the attention mechanism aims to allocate influence weights for each concept category based on student-specific attributes.

4.3. Diagnosis interaction module

Prior to conducting a diagnostic analysis, we extracted relevant attributes from the trainable matrix. In particular, student attributes encompass skill and speed, whereas exercise attributes include difficulty and discrimination. To obtain the characteristic attributes of the students, we trained the output of the attention module in the higher-order cognitive mechanism using skill and speed attributes. Similarly, a comparable approach was employed to derive the final exercise characteristic attributes. As shown in Fig. 5(c), the specific formula for the diagnostic interaction module is expressed as follows:

$$\theta = x_t' \odot (v_i \times t_i - \tau_j) - b_j \tag{11}$$

where x_t', v_i, and t_i denote the final characteristic attributes, the response speed and response time of the student. τ_j and b_j denote the workload attribute and difficulty attribute of exercise.

To predict the student response results, we employ fully connected neural networks with a sigmoid activation function. The outputs derived from the diagnosis interaction module serve as inputs for the predictors:

$$f_1 = \sigma\left(W_1 \times \theta^T + b_1\right) \tag{12}$$

$$f_2 = \sigma\left(W_2 \times f_1 + b_2\right) \tag{13}$$

$$f_3 = \sigma\left(W_3 \times f_2 + b_3\right) \tag{14}$$

where $\hat{y}_i$ is the probability of a student answering correctly based on a predictor, and σ is the activation function.

To enhance model interpretability, we adopted the widely-used monotonicity hypothesis (Rosenbaum, 1984) from IRT, which posits that a student's score on an exercise increases with their proficiency in the underlying concepts. To satisfy this hypothesis during training, we imposed constraints on each element of W_1, W_2, and W_3 to ensure positivity.

4.4. Model training module

As in prior cognitive diagnosis studies, we formalized student performance prediction as a dichotomous (right or wrong) task and employed the binary cross-entropy (BCE) loss as the basis for the loss function. BCE is commonly used in binary classification tasks, particularly when predicting probabilities for binary outcomes. The loss function of the HO-NCD model is the cross-entropy between the output $\hat{y}_i$ and true label y_i:

$$loss = -\sum_i \left(y_i log \hat{y}_i + \left(1 - y_i\right) log\left(1 - \hat{y}_i\right)\right) \tag{15}$$

Algorithm 1 The HO-NCD for Students' Performance Prediction.

Input: The trainable embedding matrices $\mathbf{A}, \mathbf{V}, \mathbf{C}, \mathbf{D}$;
The exercise-concept correlation matrix $\mathbf{Q}$;
The response logs set $\varepsilon = \{(i, j, y_i)\}$;
The concept number U, concept categories G;
The one-hot vector of student $\mathbf{h}_i^s$;
The one-hot vector of exercise $\mathbf{h}_j^p$;
The batch size b;
The max number of epochs N.
Output: $\mathbf{A}$: Student's cognitive states; $\mathbf{Y}$: Student's predicted scores.

Initialize the model parameters.
for $n \leftarrow 1$ **to** N **do**
  for $\varepsilon_{batch} \leftarrow$ sample(ε, b) **do**
    for each student $i \in \varepsilon_{batch}$ **do**
      $\alpha_i \leftarrow \mathbf{h}_i^s \times \mathbf{A}$, $\nu_i \leftarrow \mathbf{h}_i^s \times \mathbf{V}$ {Student feature embedding}
      $b_j \leftarrow \mathbf{h}_j^P \times \mathbf{C}$, $\tau_j \leftarrow \mathbf{h}_j^P \times \mathbf{D}$ {Exercise feature embedding}
      $C = \{C_{l1}, C_{l2}, \ldots, C_{lG}\} \leftarrow \{C_1, C_2, \ldots, C_U\}$ {Concept feature embedding through manual sorting}
    end for
    for each student $(i, j) \in \varepsilon_{batch}$ **do**
      $\sigma_l \leftarrow \sum_{g=1}^{G} w_g \times c_{lG}^k$ {Compute concept-level aggregation}
      $\sigma_l' \leftarrow \sum_{u=1}^{U} w_u \times \sigma_u$ {Aggregate lower-order abilities}
      $\theta \leftarrow x_t' \odot (v_i \times t_i - \tau_j) - b_j$ {Compute attention-enhanced prediction}
      Update σ_l with c_{lG}^k and Attention weights w_g.
      Calculate θ with $x_t', v_i, t_i, \tau_j, b_j$.
      $\hat{y}_i \leftarrow$ MLP(θ^T) {Predicted score}
      $\mathcal{L} \leftarrow$ CrossEntropyLoss$(y_i, \hat{y}_i)$ {Compute the loss}
    end for
  end for
  return $\mathbf{A}$ and $\mathbf{Y}$
end for

where $\hat{y}_i$ is the probability of a student answering exercises correctly from predictor.

During model training, the trainable parameters are continuously updated through backpropagation in the neural network. The students' cognitive attributes are obtained upon completion of training. The pseudocode is shown in Algorithm 1.

5. Experiments

First, we compared our HO-NCD model with several baseline approaches in the context of predicting student performance. We conducted an interpretive evaluation of these models.

5.1. Dataset description

We used three public datasets compiled with real data: Junyi, ASSIST2017, and PISA2015.

Junyi: Junyi contains data derived from the online learning platform of Junyi Academy. The original Junyi dataset includes more than 20 million response records distributed across different learning sessions. Keeping with the static assumption of cognitive diagnostics, we analyzed the relationship between student response data and the included exercises and concepts.

ASSIST2017: ASSIST2017 is sourced from the "2017 ASSISTments Datamining Competition" and provides records of students' answers from 2004 to 2007, as well as relationships between the included exercises and concepts.

PISA2015: PISA is the world's definitive online exercise dataset composed of high-quality exercises and a record of students' response

Table 2
Dataset statistics.

Statistics	Junyi	ASSIST2017	PISA2015
Students	15 000	1709	6000
Exercises	718	102	17
Skill	39	3162	11
Concepts per exercise	1.1	1.19	1.28
Proportion of correct response	0.76	0.44	0.52

results and times. The statistics of the datasets used in this study are listed in Table 2.

To cluster the knowledge points in the dataset, we employ manual clustering. Specifically, based on the new curriculum, we divide the knowledge concepts into unequally sized groups according to their correlations, ensuring they align with the dataset. For example, the PISA dataset contains 11 knowledge concepts, which are categorized into three groups based on their intercorrelations.

As a preprocessing step, we eliminate any duplicate responses from the same students to the same exercise, retaining only the initial records. In addition, to ensure an adequate number of response records for diagnostic purposes, we exclusively include students who completed more than 15 exercises (Wang et al., 2020). For each student, we randomly divide all datasets into an 80%/20% ratio for training and testing respectively. Consequently, both groups perform an equal number of exercises but with different contents. The training set is used for model training and hyperparameter tuning and the exercise set is used for model evaluation. These hyperparameters are typically tuned using a combination of grid search and cross-validation, including the embedding dimension, learning rate, and others. Specifically, the embedding dimension d for student, exercise, and concept vectors is selected by evaluating model performance across a range of values, such as $d \in \{16, 32, 64, 128\}$. A smaller dimension may fail to capture the complexity of the relationships, while a larger dimension could lead to overfitting and increased training times. Additionally, the learning rate is tuned using a logarithmic search over a range of values $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$. The optimal learning rate is selected based on the validation loss, ensuring stable training without premature convergence or excessive fluctuations. All models are repeatedly trained and tested from scratch, and their performance was evaluated via averaging.

5.2. Baselines

To evaluate the proposed HO-NCD model's performance, we compared it with cognitive diagnosis models based on statistical methods (i.e., IRT, MIRT, and DINA), CDMs based on neural networks (i.e., NCD, ICD, ENCD, and MFNCD), and higher-order CDMs (i.e., fuzzy CD and Hier NCD). Detailed descriptions of the baselines are provided as follows:

DINA (De La Torre, 2009): DINA implements the micro-diagnosis of students through Q-matrix, which allows access to the student's mastery of various skills and provides targeted assistance in improving weaknesses.

IRT (Lord, 2012): IRT is the one of classical cognitive diagnosis models. This method models student and exercise attributes through logistic functions. The diagnosis result of it is a comprehensive but ambiguous ability value.

MIRT (Ackerman et al., 2003): MIRT is proposed to better model the relationship between the multiple abilities, exercise characteristics, and probability of answering correctly that subjects need to complete exercises.

Fuzzy CD (Wu et al., 2015): Fuzzy CD uses a model inspired by fuzzy systems (Fuzzy Cognitive Diagnostic Framework) to address the problem of answering subjective exercises partially correctly.

Neural CD (Wang et al., 2020): This method is a deep learning-based cognitive diagnostic model that incorporates IRT to diagnose students' cognitive attributes and exercises.

Hier CDF (Li et al., 2022): This method designs a cognitive diagnostic model adapter that converts students' cognitive states and exercise characteristics into input characteristics for existing diagnostic models. Hier CDF can be implemented in combination with a variety of diagnostic models (e.g., IRT, MIRT, etc.) and has broad applicability.

ICD (Qi et al., 2023): ICD is a new cognitive diagnosis model based on neural network, which considers the interactions between concepts and the quantitative relationship between exercises and concepts.

ENCD (Huang, Geng, Yang, Hu, Chen et al., 2024): ENDC is a simplified version of the LS-ENCD that models the interactions between student attributes and exercise features. It also considers slipping and guessing, thereby enhancing the diagnosis accuracy of cognitive assessment.

MFNCD (Huang, Geng, Yang, Hu, Ou et al., 2024): MFNCD integrates multidimensional features by using students' reaction times as process information. This facilitates the simultaneous modeling of students' reaction accuracy and reaction speed using neural networks.

5.3. Evaluation metrics and environment setting

We employed diverse metrics from various perspectives to evaluate the performance of our model. Given that the exercises in our datasets were objective, with scores of 0 or 1 representing incorrect or correct answers, respectively, both categorical and regression metrics were used in this study. On the categorical side, we used prediction accuracy (ACC) and area under the curve (AUC) to assess predictive performance. On the regression side, the root mean square error (RMSE) was used to measure the gap between predicted and actual scores.

In addition, the hyperparameter learning rate was set to 0.002, and the hidden dimensions were set to 16. The dimensions of the fully connected layer were defined as 512, 256, and 1. All models were implemented using Python 3.8 on a Linux server equipped with a 3.5 GHz Intel (R) Core (TM) i9-10920X CPU and an RTX 3090 GPU, with a total running memory of 24 GB. All experiments were repeated five times, and the average results from these repetitions were considered the final experimental outcomes.

5.4. Experimental results and analysis

The direct assessment of model performance using diagnosed cognitive states is limited by the scarcity of available cognitive states in existing datasets (Wu et al., 2015). Therefore, most prior studies indirectly evaluated model performance by predicting students' response scores. To evaluate the HO-NCD model's performance in predicting students' response scores, extensive experiments were conducted on three authentic datasets, with results compared between the proposed and baseline models.

5.4.1. Student performance prediction

We divide the student's response prediction into three categories: CDMs based on statistical methods, those based on neural networks, and higher-order CDMs.

Table 3 presents the predictive performance and assessment metrics for all models, including accuracy (ACC), root mean square error (RMSE), and area under the curve (AUC), with the best results highlighted in bold. Notably, because of the specific nature of the fuzzy CD model, which focuses on diagnosing students' subjective answers to exercises, the model deviates from the characteristics of other baseline models' datasets. Therefore, we modified the framework of the fuzzy CD model by excluding the modeling of subjective exercises and considering only objective exercise answers provided by students. Additionally, we incorporated ACC and AUC as evaluation metrics to further analyze our model's effectiveness. As depicted in Table 3, our model outperformed the baselines across all datasets used. Specifically, our hierarchical attention network approach significantly enhanced prediction performance compared to traditional models, thus validating its efficacy in predicting student responses.

Table 3
Results on student performance prediction.

Model		Junyi			PISA2015			ASSIST2017		
		RMSE	ACC	AUC	RMSE	ACC	AUC	RMSE	ACC	AUC
CDMs based on statistical methods	IRT	0.382	0.796	0.744	0.664	0.559	0.566	0.493	0.595	0.656
	DINA	0.439	0.692	0.707	0.591	0.652	0.651	0.549	0.550	0.598
	MIRT	0.390	0.796	0.752	0.452	0.688	0.759	0.554	0.629	0.668
CDMs based on neural networks	NCD	0.436	0.713	0.783	0.474	0.691	0.747	0.442	0.704	0.770
	ICD	0.382	0.802	0.804	0.523	0.661	0.745	0.433	0.717	0.771
	ENCD	0.365	0.812	0.821	0.465	0.706	0.767	0.445	0.711	0.755
	MFNCD	0.364	0.817	0.814	0.455	0.708	0.775	0.435	0.715	0.775
Higher-order CDMs	Fuzzy CD	0.375	0.797	0.755	0.445	0.677	0.712	0.452	0.684	0.735
	Hier NCD	0.401	0.751	0.757	0.455	0.692	0.781	0.443	0.688	0.780
Our method	**HO-NCD**	**0.349**	**0.833**	**0.835**	**0.441**	**0.711**	**0.785**	**0.432**	**0.719**	**0.787**

Table 4
Results of ablation experiment.

	A	B	C	D	Junyi			PISA2015			ASSIST2017		
					RMSE	ACC	AUC	RMSE	ACC	AUC	RMSE	ACC	AUC
HO-NCD-A	–	✓	✓	✓	0.362	0.820	0.778	0.447	0.700	0.774	0.444	0.689	0.774
HO-NCD-B	✓	–	✓	✓	0.363	0.821	0.797	0.441	0.710	0.779	0.439	0.706	0.778
HO-NCD-C	✓	✓	–	✓	0.362	0.821	0.797	0.442	0.710	0.778	0.433	0.713	0.782
HO-NCD-D	✓	✓	✓	–	0.362	0.815	0.791	0.441	0.708	0.775	0.441	0.703	0.776
HO-NCD	✓	✓	✓	✓	0.349	0.833	0.835	0.441	0.711	0.785	0.432	0.719	0.787

5.4.2. Ablation experiments

To further verify the effectiveness of the individual network layers within the proposed HO-NCD framework, we constructed four variants of the HO-NCD model and performed ablation experiments on these configurations with different datasets. These experiments aim to systematically assess the contribution of each component to the overall performance of the model. We construct four primary variants of the HO-NCD model, in which specific network layers or key features are selectively removed or replaced while ensuring that the input–output dimensions remain consistent across all variants. This allows for the isolation of the effects of each layer while maintaining the overall structure of the model. For clarity, these models are denoted by prefixing a particular network layer with a "–" symbol to indicate the replacement or removal of the corresponding network layer or key features. The details of each specific control model are provided below.

Concept State Filtering Layer: In this variant, the concept state filtering layer of HO-NCD, responsible for refining the interaction between concepts, is entirely removed. This variant is labeled as HO-NCD-A. By eliminating this layer, we aim to examine the impact of its absence on the model's ability to accurately diagnose cognitive states.

Dot-Attention: In this variant, the higher-order interaction layer, which utilizes a hierarchical attention mechanism to capture complex relationships, is replaced with a simple summation averaging approach. This simplification effectively removes the attention mechanism and assumes equal weighting for all cognitive states. This variant is labeled as HO-NCD-B. The goal is to assess the significance of the hierarchical dot-attention mechanism in enhancing the model's ability to identify and interpret higher-order interactions within the cognitive diagnosis framework.

Aggregation Layer: The aggregation layer is replaced with a simple summation averaging treatment denoted HO-NCD-C. By removing the more sophisticated aggregation mechanism, we aim to evaluate the impact of a simpler approach to combining features and its effect on overall model performance.

Diagnosis Interaction Module: In this variant, we remove the final characteristic attributes of the student, which is labeled as HO-NCD-D. By removing the final output of the higher-order cognitive structure attention module, we aim to assess the impact of this module on the overall model performance.

Table 4 reports the results of each case. The following conclusions can be drawn from the table. Firstly, regardless of whether the layer is removed or replaced with a plain method, the effectiveness of the model is reduced. Secondly, among them, HO-NCD-D has the worst results, which indicates the importance of the Higher-Order Cognitive Structure Attention module in the overall model. This observation holds true across different datasets and various model setups. These findings highlight that the hierarchical attention network plays a pivotal role in the modeling process of the proposed approach.

5.4.3. Degree of agreement analysis

To assess the interpretability of the HO-NCD model — that is, the plausibility of its diagnostic results — we conducted a monotonicity analysis (DOA) (Wang et al., 2020). In essence, if a student demonstrates greater proficiency in concept c_k than s_n, denoted by $\alpha_i > \alpha_n$, then their performance on exercise e_j containing c_k should surpass that on exercise e_j, indicated by $r_i > r_n$. Specifically, for the concept c_k, $DOA(k)$ is calculated as follows:

$$DOA(k) = \frac{1}{Z}\sum_{a=1}^{N}\sum_{h=1}^{N} I(F_{ak} > F_{hk}) \times H$$

where $H = \frac{\sum_{j=1}^{M} I(Q_{jk}=1)\wedge J(j,a,b)\wedge I(r_{aj}>r_{bj})}{\sum_{j=1}^{M} I(Q_{jk}=1)\wedge J(j,a,b)\wedge I(r_{aj}\neq r_{bj})}$, $Z = \sum_{a=1}^{N}\sum_{b=1}^{N}\delta(F_{ak}^{s}, F_{bk}^{s})$.

We averaged $DOA(k)$ across all concepts to evaluate the quality of the diagnostic results (i.e., the proficiency of concepts gained through the model). In the traditional models, the comparison was conducted solely with DINA, as the IRT and MIRT models do not show a clear correspondence between their student characteristics and concepts. The monotonicity of HO-NCD was analyzed using SPSS software and Python calculations. By comparing the DOA results obtained from these approaches, the results obtained with the SPSS analysis were found to be superior when analyzing the same dataset. Upon reviewing the specific comparison information, we have found that human-written Python statements can introduce subjective errors that negatively affect the analysis results. Consequently, SPSS was selected as the preferred tool for result analysis. Fig. 8 illustrates the DOA analysis outcomes, where HO-NCD can be observed to exhibit significantly higher values than the baselines, thereby demonstrating our model's superior interpretability.

5.4.4. Weighting analysis of high-level hierarchical relationships

Taking the PISA2015 dataset as an illustrative example, we initially established adaptive weights w_i between concepts and abilities. To further investigate the relationship between the concepts and students'

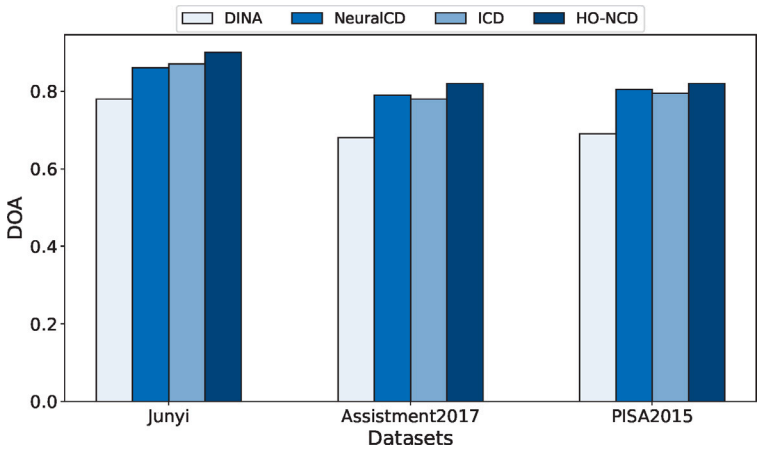


Fig. 8. Results of DOA analysis. The DOAs of HONCD on the three datasets is significantly higher than the other baselines, which proves that our method is more interpretable.

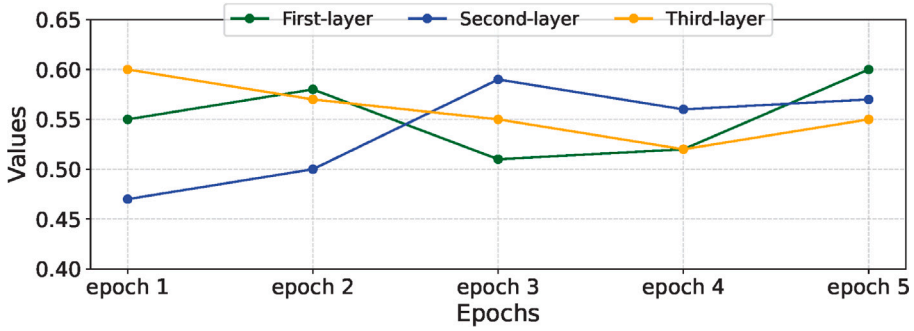


Fig. 9. Distribution of influence weights. For different attention network layers, HO-NCD learned the w_i's values generally in the range of [0.5–0.6].

skills, we analyzed and visualized the distribution of numerical values representing the influence weights acquired through adaptations learned in different network layers. As shown in Fig. 9, the w_i's values learned by the HO-NCD for different attention network layers generally fall within the range of [0.5–0.6]. The slight variations observed among them could potentially be attributed to the varying complexities involved in parameter learning across different neural network layers.

Subsequently, we analyzed the distribution of the influence weights for each layer within the attention module of the higher-order cognitive structure. As depicted in Fig. 10(a), the distribution of influence weights for all concepts in the concept state-filtering layer of HO-NCD reveals minimal disparities in weight values in the first layer, indicating a consistent level of concept mastery among most students. This finding aligns with our results from ablation experiments. Notably, highlighted weight values suggest that a majority of students possess a lower proficiency level in this particular concept, thereby emphasizing the need for teachers to prioritize its examination and focus. As shown in Fig. 10(b), the influence weights of students' skill mastery varied in the lower-order ability attention layer, predominantly hovering around 0.5. These results demonstrate that HO-NCD can effectively extract and aggregate related concepts assessed in exercises. With the assistance of the attention mechanism, it can more effectively explore the intrinsic relationship between concepts and lower-order skills. In terms of the highlighted weight values, it is plausible to prioritize teaching as most students seem to lack a high level of mastery for this particular skill.

In the higher-order ability aggregation layer, utilizing the PISA2015 dataset as an illustration, we aggregate diverse skills to generate the ultimate higher-order trait (e.g., math literacy) for the HO-NCD model.

5.4.5. Case study

The assessment of students' conceptual mastery is a crucial cognitive diagnostic process that can inform targeted learning strategies. Fig. 11(a) displays the extracted skill mastery data from PISA2015, with each value representing a student's level of proficiency in a specific concept. Higher values indicate greater mastery. As shown in Fig. 11(b). We averaged the students' skill mastery. The outward to inward values in each layer indicate the average mastery of concepts, skills, and higher-order abilities of the students, respectively.

Table 5 presents examples of diagnosis results for individual students. When performing diagnosis analyses with HO-NCD, in addition to obtaining diagnosis analyses for all attributes, it is also possible to obtain predicted values for the student on higher-order traits (e.g., mathematical literacy) as well as hierarchical latent traits (e.g., arithmetic, reasoning, and spatial concepts, etc.). In the case of students s_{14} and s_{30}, for example, even though the two students' responses are exactly the same, they differ in their performance on underlying characteristics, indicating a disparity in their mastery of skills. By comparing the skills mastery status of students s_{92} and s_{289}, it becomes evident that the depth of concept understanding influences their latent traits and higher-order characteristics. For example, student s_{289} has mastered all the concepts. Therefore, this student has higher latent and higher-order traits than the other students. Furthermore, an observation can be made from the students' responses that when student a answers more exercises correctly than student b, student a naturally possesses superior higher-order traits than b, which aligns with the aforementioned monotonicity assumption. Specifically, when student s_{59} answers significantly fewer exercises correctly than student s_{171}, it corresponds to higher levels of all latent traits α as well as higher-order traits x_t. In conclusion, based on the analysis results from PISA2015 data set, it is evident that HO-NCD demonstrates robust validity and interpretability.

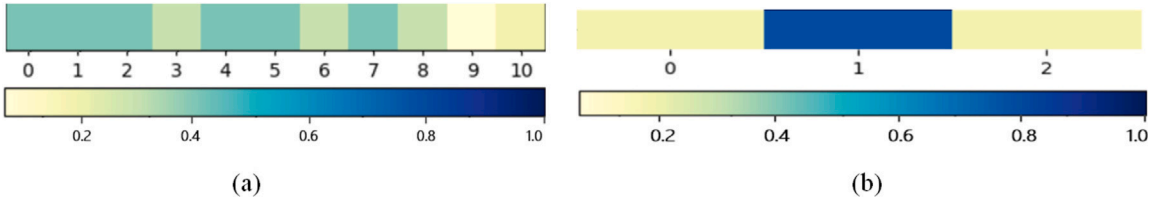


Fig. 10. Distribution of influence weights. (a) The distribution of influence weights for all concepts; (b) the influence weights of students' skill mastery. The horizontal axis represents the number of skills, where eleven knowledge points are represented in (a) and three lower-order skills are represented in (b). The lower color code indicates the magnitude of the influence weight (0–1).

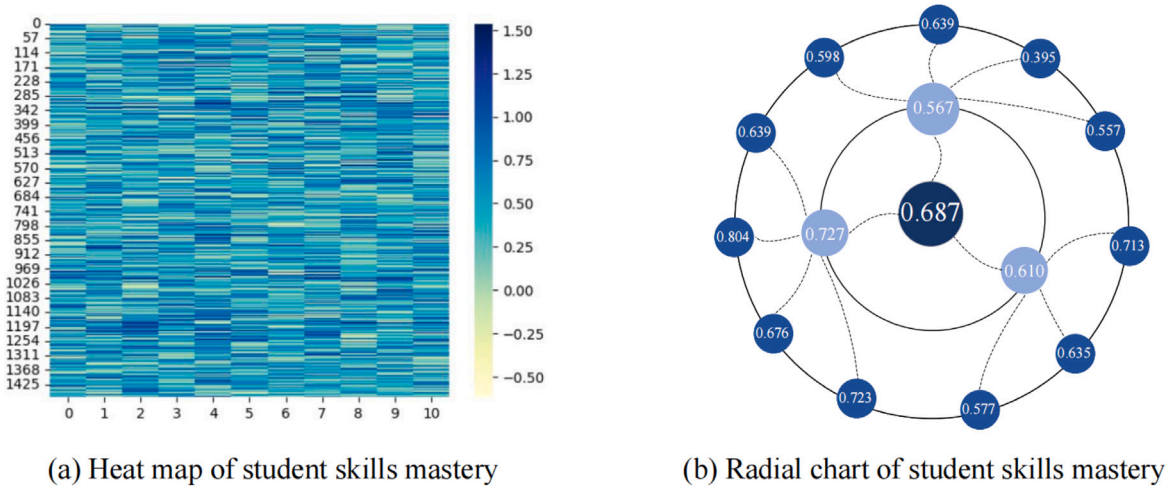


Fig. 11. Case study.

Table 5
Examples of diagnosis results for individual students.

Student	Skill mastery	α_1	α_2	α_3	x_t
s_2	11100010110	0.514	0.713	0.697	0.641
s_{14}	10011111001	0.515	0.637	0.676	0.609
s_{20}	11101010101	0.727	0.639	0.702	0.689
s_{30}	10011111001	0.527	0.635	0.779	0.647
s_{59}	00100100011	0.325	0.452	0.395	0.391
s_{92}	00000000000	0.114	0.156	0.289	0.186
s_{171}	11111011111	0.748	0.713	0.720	0.727
s_{289}	11111111111	0.885	0.858	0.774	0.839

6. Conclusion

This paper proposes a cognitive diagnosis model named HO-NCD based on hierarchical attention networks to mine the potential relationship between concepts examined in exercises and students' abilities. Specifically, we commence by discussing the underlying factors related to students and exercises while adhering to the monotonicity assumption for interpretability purposes. Subsequently, we implement a specific higher-order cognitive diagnosis model within this framework, demonstrating its feasibility by deriving students' skill mastery using a three-layer attention network.

Our approach offers two key advantages. Firstly, it delves deeply into the internal relationship between concepts and their potential connection with students' abilities; secondly, it constructs hierarchical attention networks that outperform traditional models. Extensive experiments conducted on real datasets validate the superiority and interpretability of HO-NCD in mining potential relationships for cognitive diagnosis while providing comprehensive and interpretable results. This work is expected to offer a fresh perspective on cognitive diagnosis and to stimulate future advancements. However, there are some limitations that need to be further strengthened in future studies. On the one hand, our model demonstrated strong performance on the dataset used in our experiments. However, further research is needed to assess its scalability when applied to larger, more complex datasets. On the other hand, our model primarily relied on outcome data and did not take into account process data, such as students' interaction histories or time-series data. Future work could explore the integration of such process-oriented data to uncover deeper insights into students' learning behaviors and performance. Building on this, future work could investigate deeper relationships between students and exercises, such as uncovering causal relationships to better understand the influence of specific concepts on student performance. Additionally, future work may incorporate more process-oriented data, such as students' interaction histories or time-series data, to further enhance the accuracy of cognitive diagnosis.

CRediT authorship contribution statement

Tao Huang: Conceptualization, Methodology, Supervision, Writing – original draft. **Yuxia Chen:** Methodology, Validation, Formal analysis, Writing – original draft. **Jing Geng:** Methodology, Writing – original draft, Writing – review & editing. **Huali Yang:** Investigation, Resources, Writing – original draft. **Shengze Hu:** Visualization, Validation, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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